

DAT12-1

Introduction to Data with Spreadsheet

Observing, organising, and describing data

Look at the data before you touch it.

Albert School — 16 hours

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1 The spreadsheet environment

A spreadsheet organises data into a grid of cells addressed by column letter and row number (for example B4). A cell holds a value (a number, a date, or text) or a formula. The first task of data work is to enter, edit, and format data correctly so that the values mean what they appear to mean.

Definition 1 A cell's **type** is how the spreadsheet interprets its content: number, date, or text. Formatting controls how a value is displayed (decimal places, currency, date format) without changing the underlying value.

Correct entry means each column holds one kind of value, numbers are stored as numbers rather than text, and dates are stored as dates. Editing a cell replaces or amends its content; formatting changes its display.

Common pitfall A number stored as text will not be summed or averaged correctly, and a date stored as text will not sort chronologically. Check the stored type, not only the displayed value.

2 Core functions

A formula begins with = and computes a result from values or cell references. The following functions answer simple data questions, each appropriate to a different situation.

Definition 2

- **SUM(range)** adds the numbers in a range.
- **AVERAGE(range)** returns their arithmetic mean.
- **IF(condition, value-if-true, value-if-false)** returns one of two values depending on whether a condition holds.
- **COUNTIF(range, criterion)** counts the cells in a range that meet a criterion.
- **VLOOKUP(key, table, column, FALSE)** finds a row by its key in the first column of a table and returns a value from another column of that row.
- **INDEX(range, row, column)** returns the value at a position; **MATCH(key, range, 0)** returns the position of a key. Combined, **INDEX/MATCH** performs a lookup without requiring the key to be in the first column.

SUM and **AVERAGE** summarise a column of numbers. **IF** classifies or branches on a condition. **COUNTIF** counts occurrences. A lookup retrieves a value associated with a key from another table; **VLOOKUP**

requires the key in the first column and returns a column to its right, while `INDEX/MATCH` has neither restriction.

3 Sorting, filtering, and grouping

Sorting reorders rows by the values in one or more columns. Filtering hides rows that do not meet a stated condition, leaving the rest in view. Grouping aggregates rows that share a value in some column and reports a summary (count, sum, average) per group.

These operations surface patterns and anomalies: sorting brings the largest or smallest values together, filtering isolates the rows of interest, and grouping reveals how a quantity varies across categories.

4 Pivot tables

Definition 3 A **pivot table** summarises a table by grouping rows according to one or more fields and computing an aggregate (count, sum, average) of another field for each group. Rows and columns of the pivot table correspond to the values of the grouping fields; the body holds the aggregates.

Building a pivot table means choosing which fields define the rows and columns and which field is aggregated, and by which function. Interpreting it means reading the aggregates back as a statement about the data, and explaining that statement in business terms — what the numbers say about the question at hand.

5 Data quality at the spreadsheet level

Data quality issues are visible at the spreadsheet level and can be identified and corrected manually.

Definition 4

- **Duplicates:** rows that repeat the same record, inflating counts and sums.
- **Inconsistent formats:** the same value entered in different ways (for example differing date formats, capitalisation, or units), so that equal values are not recognised as equal.
- **Missing values:** cells left empty where a value is expected.

Manual correction means removing or merging duplicates, standardising formats so equal values match, and deciding for each missing value how to handle it. Identifying these issues is a matter of inspecting the data — sorting, filtering, and counting reveal repeats, mismatches, and blanks.

6 Visualization

Three chart types each suit a different kind of question.

Definition 5

- A **bar chart** compares a quantity across categories.
- A **line chart** shows how a quantity changes over an ordered axis, typically time.
- A **scatter plot** shows the relationship between two numeric variables, one on each axis.

Choosing the right chart means matching the chart to the question: comparison across categories calls for a bar chart, change over time for a line chart, and the relationship between two variables

for a scatter plot. A chart that does not match its question misleads as surely as a wrong calculation.

Common pitfall A misleading chart is as dangerous as a wrong model. A chart that suits the data but not the question, or whose axes distort the values, communicates a false conclusion.

7 From business question to spreadsheet answer

Answering a business question with a spreadsheet has three steps: formulate the question precisely, select the operations that answer it (functions, sorting, filtering, grouping, pivot tables, charts), and interpret the result back in terms of the original question.

The discipline is to choose only the operations the question requires and to state what the result means, rather than producing numbers without an answer.

8 Documentation

Definition 6 A spreadsheet analysis is **reproducible** when another person can follow it, understand each step, and arrive at the same result without assistance.

Documentation records the source of the data, the steps taken, the assumptions made (for example how missing values were handled), and where each result comes from. An analysis is complete only when it is documented well enough to be reproduced by someone who was not present when it was made.

Exercises

1. Enter a small dataset, ensuring numbers are stored as numbers and dates as dates, and format it for correct display. Identify any value stored as the wrong type and correct it.
2. Using SUM, AVERAGE, IF, COUNTIF, and a lookup (VLOOKUP or INDEX/MATCH), answer a set of simple questions about a dataset. For each, state why that function is the appropriate one.
3. Sort, filter, and group a table to surface a pattern and an anomaly. Describe what each operation revealed.
4. Build a pivot table from a dataset, then interpret its output in business terms.
5. Find the duplicates, inconsistent formats, and missing values in a dataset and correct each manually, recording what you changed.
6. Construct a bar chart, a line chart, and a scatter plot from appropriate data, and explain for each why that chart type fits the question it answers.
7. Formulate a business question, answer it using only spreadsheet operations, and interpret the result.
8. Document one of the analyses above so that another person can reproduce it without assistance.